

RUIYANG WANG

<https://ruiyangwangw.github.io>

Robotics & LLM Research | Embodied AI | Multi-Robot Autonomy

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EDUCATION

- **Duke University** Durham, NC
Ph.D. in Electrical and Computer Engineering *Aug 2024 – Apr 2028 (Expected)*
- **University of Michigan, Ann Arbor** Ann Arbor, MI
M.S. in Robotics; B.S. in Mechanical Engineering *Aug 2018 – Dec 2023*

RESEARCH EXPERIENCE

- **Duke University — Cyber-Physical Systems Lab** Durham, NC
Ph.D. Researcher / Advisor: Prof. Miroslav Pajic *Aug 2024 – Present*
 - Developed **multi-robot autonomy** methods using LLM/VLMs for exploration and search coordination, **graph attention networks** for trust inference, and **neural-symbolic reasoning** for deadlock resolution.
 - Building **graph-based planning harnesses** to verify and repair LLM/VLM-generated robot action plans, improve **long-horizon task completion**, and check task and safety constraints before execution.

SELECTED PUBLICATIONS & MANUSCRIPTS

- **GAVEL: Graph World Models for Verified and Efficient Long-Horizon LLM Task Planning**
Ruiyang Wang, Hao-Lun Hsu, Swarajh Mehta, Jiwoo Kim, Zhihao Dou, Miroslav Pajic
Submitted to IEEE International Conference on Robotics and Automation (ICRA), 2027
 - Designed graph-based plan verification, repair, and belief-aware task ordering; raised Qwen3-8B success from **41.2%** to **91.8%** on single tasks and **19.9%** to **92.6%** on multi-task instructions in OmniGibson with BEHAVIOR-1K scenes.
- **COMRES-VLM: Coordinated Multi-Robot Exploration and Search using Vision Language Models**
Ruiyang Wang, Hao-Lun Hsu, David Hunt, Jiwoo Kim, Shaocheng Luo, Miroslav Pajic
Submitted to IEEE Robotics and Automation Letters (RA-L)
 - Built a **VLM** coordinator using **natural-language hints**; improved search efficiency by **47%** over Hungarian and **55%** over Co-NavGPT in large simulated environments, and evaluated on two **Unitree Go2** robots in real world experiments.
- **NeST-Bayes — Who to Trust in Multi-Robot Systems: Joint Robot and Object Hypothesis Trust Inference via Neural-Symbolic Cross-Validation**
Ruiyang Wang, Hao-Lun Hsu, Haocheng Meng, Songqiao Xie, Miroslav Pajic
Submitted to IEEE Robotics and Automation Letters (RA-L)
 - Developed **joint robot and object trust inference** from cross-robot report consistency using **heterogeneous graph attention** and temporal Beta updates; achieved **96.4%/97.0% robot/object precision** against aggressive optimized attacks and **97.3%/97.7%** on simulated Spot robots in Webots without retraining.
- **SAGR — Semantic Area Graph Reasoning for Multi-Robot Language-Guided Search**
Ruiyang Wang, Hao-Lun Hsu, Jiwoo Kim, Miroslav Pajic
arXiv preprint
 - Designed **semantic area graphs** for LLM coordination with prompts of **fewer than 500 tokens**; reduced large-environment search time by **18.8%** versus AEP+DVC using GPT-4o, evaluated on 100 HM3D scenarios.
- **NSDR — Neuro-Symbolic Deadlock Resolution in Multi-Robot Systems**
Ruiyang Wang, Bowen He, Miroslav Pajic
Accepted at Learning for Dynamics and Control (L4DC), 2025
 - Learned which robots should yield using **Neural Logic Machines** and MPC feasibility checks; generalized from **2–5 to 8 and 10 robots**, reducing average arrival time by **23% versus greedy** in the 10-robot test.

TECHNICAL SKILLS

Programming Languages: Python, C++, MATLAB, R, Julia

ML & LLMs: PyTorch, Hugging Face Transformers, scikit-learn, SFT, LoRA, RAG, vLLM, Weights & Biases

LLM Agents: Planning harnesses, structured action generation, plan verification and repair

Systems & Cloud: Linux (Ubuntu), Git, Docker, ROS2, AWS, Google Cloud

Simulation Tools: OmniGibson, Webots, PyBullet, HM3D, OpenAI Gym, BEHAVIOR-1K